CAN Bus Fundamentals: Overview and physical layer.

The Controller Area Network (CAN) protocol was developed by Bosch and introduced in 1986. It was created so that microcontrollers and devices inside a vehicle could communicate without a host computer. CAN is standardised as ISO 11898.

CAN is a multi-master, message-based broadcast bus: every node sees every message, and any node may start transmitting when the bus is idle. Messages are identified by content rather than by the address of a destination node.

On the physical layer CAN uses differential signalling on a twisted pair of wires called CAN_H and CAN_L. The difference between the two lines carries the bit value, which makes the bus robust against electromagnetic noise. A 120 ohm termination resistor is placed at each end of the bus to prevent signal reflections. Without correct termination, reflections corrupt the waveform and cause errors at higher bit rates.

The bus has two logical states. The dominant state represents a logical 0 and the recessive state represents a logical 1. When one node drives the bus dominant while another drives it recessive, the bus ends up dominant.

CAN Bus Fundamentals: Frame format and arbitration.

A standard CAN frame has an 11-bit identifier. The extended frame format has a 29-bit identifier. A classical CAN data field carries between 0 and 8 bytes. Each frame also contains a 15-bit CRC field for error detection and an acknowledge slot in which receivers confirm correct reception.

When two nodes begin transmitting at the same moment, CAN resolves the conflict through non-destructive bitwise arbitration. Each transmitter monitors the bus while sending its identifier. If a node sends a recessive bit but reads a dominant bit, it has lost arbitration, stops transmitting and becomes a receiver. The message with the lower numerical identifier wins because it contains dominant bits earlier, so a lower identifier means a higher priority. The winning message is not damaged and no time is lost.

To keep receivers synchronised, CAN uses bit stuffing: after five consecutive bits of the same polarity the transmitter inserts a bit of the opposite polarity, which the receivers remove again.

CAN Bus Fundamentals: Speed, CAN FD and error handling.

Classical CAN supports bit rates up to 1 Mbit/s. The achievable bus length falls as the bit rate rises: roughly 40 m at 1 Mbit/s, about 100 m at 500 kbit/s and up to about 500 m at 125 kbit/s.

CAN FD (flexible data-rate) extends the classical protocol. A CAN FD frame can carry up to 64 data bytes instead of 8, and the data phase can run at a higher bit rate than the arbitration phase when the bit rate switch (BRS) bit is set. Data-phase rates of 5 to 8 Mbit/s are commonly quoted.

CAN has five error types: bit error, stuff error, CRC error, form error and acknowledgement error. Every node keeps a transmit error counter (TEC) and a receive error counter (REC). A node is error-active while both counters are low. When either counter exceeds 127 the node becomes error-passive, and when the transmit error counter exceeds 255 the node goes bus-off and stops participating until it is reset. This mechanism isolates a faulty node so that it cannot block the whole network.